M. Melvin Calvin, A.A. Benson and their co-workers in the year 1957 found this path way of carbon fixation. Melvin Calvin was awarded Nobel Prize for this in 1961 and this pathway named after the discoverers as **Calvin-Benson** Cycle. Dark reaction is temperature dependent and so it is also called thermo-chemical reaction.

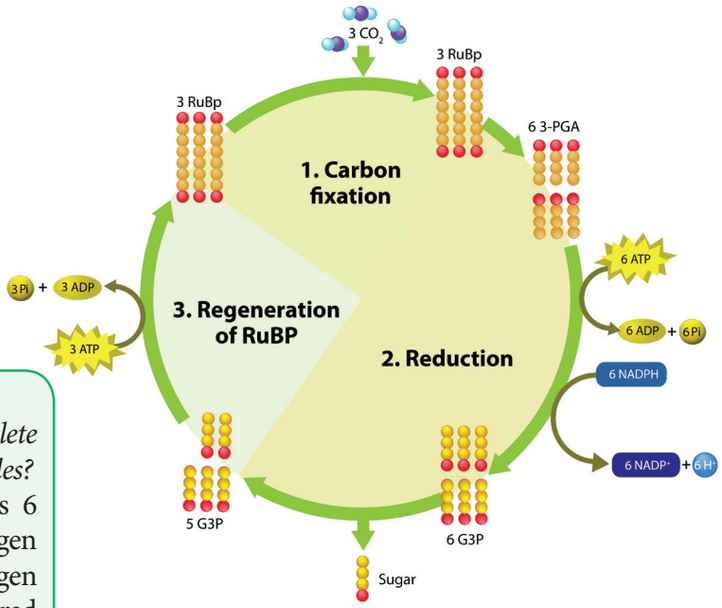


Figure 13.16: Phases of Calvin Cycle

Check your grasp!
What will be the quanta requirement for complete light reaction which releases 6 oxygen molecules?
Solution: Complete light reaction releases 6 oxygen molecules. If one molecule of oxygen evolution requires 8 quanta means, for 6 oxygen molecules $6 \times 8 = 48$ quanta of light required for complete light reaction.

Dark reaction consists of three phases: (Figure 13.16).

1. Carboxylation (fixation)
2. Reduction (Glycolytic Reversal)
3. Regeneration

Phase 1- Carboxylation (Fixation)

The acceptor molecule Ribulose 1,5 Bisphosphate (RUBP) a 5 carbon compound with the help of RUBP carboxylase oxygenase (RUBISCO) enzyme accepts one molecule of carbon dioxide to form an unstable 6 carbon compound. This 6C compound is broken down into two molecules of 3-carbon compound phospho glyceric acid (PGA) (Figure 13.17).

$$RUBP + CO_2 \xrightarrow{\text{Rubisco}} 2 \text{ molecules PGA}$$

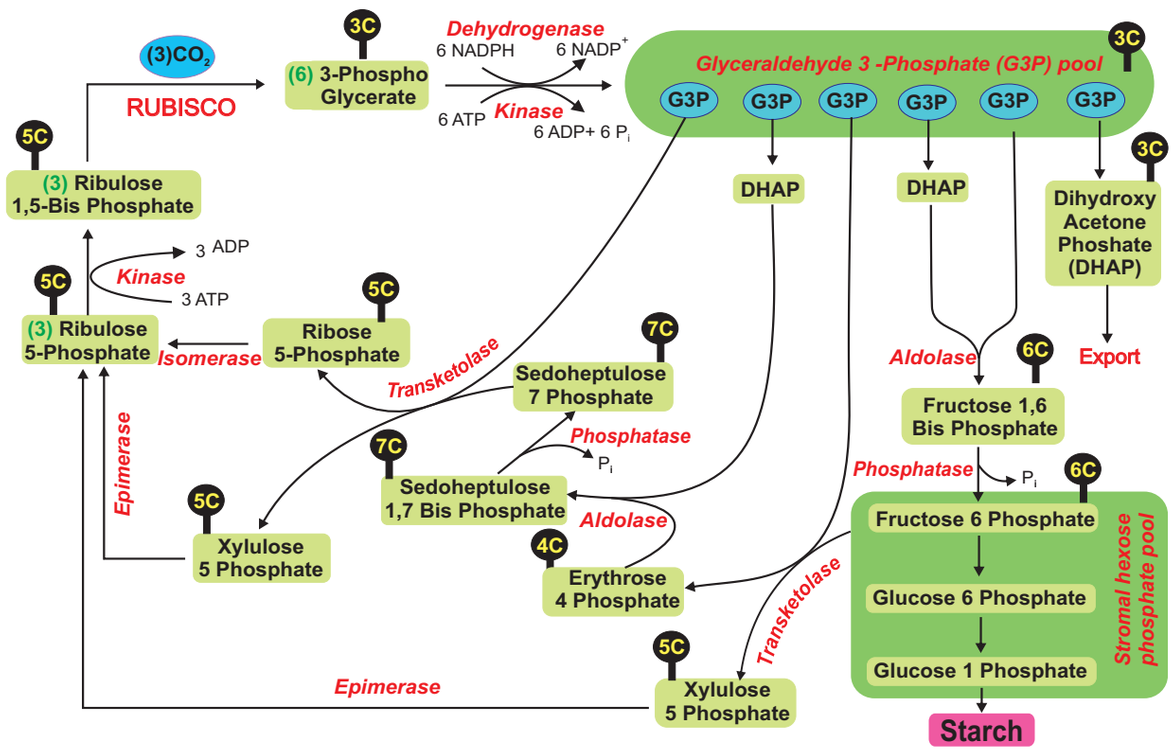


Figure 13.17: Calvin cycle

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